

Homework 6 – More about functions

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Learning outcome:

- Students should have the ability to analyze recursive functions, pointer, and reference algorithms.
- Students should be able to create multiple functions and link them together in one program.

Assignment description:

- Download this Word document and save it with your last name.
- Complete Exercise 1 and Exercise 2 in this Word document, and complete Exercise 3 in a C++ file.
- After completing all three exercises, upload the saved Word document and the C++ file.

Exercise 1)

Watch the following video about pointer and reference

[What is the Difference Between a Pointer and a Reference C++](https://www.youtube.com/watch?v=sxHng1iufQE)
<https://www.youtube.com/watch?v=sxHng1iufQE>

Follow and write the code in the video. When the video reaches 5 minutes, stop the video and write a line to print, cout, the pointer **ptr**, `cout<<"Pointer 1 = "<<ptr<<endl;`

What is printing in the console (terminal)? Ans: Pointer 1 = 0x16f98ed7c

Now, write another line to print ***ptr**, `cout<<"value 1 = "<<*ptr<<endl;`

What is printing in the console (terminal)? Ans: Value 1 = 7

Continue watching the video and continue typing the video code. When the video reaches 6 minutes, stop the video, and write a line to print the pointer **ptr**, `cout<<"Pointer 2 = "<<ptr<<endl;`

What is printing in the console (terminal)? Ans: Pointer 2 = 0x16f98ed78

Now, write another line to print ***ptr**, `cout<<"value 2 = "<<*ptr<<endl;`

What is printing in the console (terminal)? Ans: Value 2 = 21

Continue watching the video and continue typing the video code until the end of the video. When the video ends, write a line to print the reference `cout<<"ref = "<<&ref<<endl;`

What is printing in the console (terminal)? Ans: ref = 0x16fc66d7c

Exercise 2) recursive function

Given the following code

```
#include<iostream>
using namespace std;

int rose(int n){
    if (n<=0){
        return 1;
    }
    else{
        return (rose(n-1)*n);
    }
}

int main(){
    int num;
    cout<<"Enter a number: ";
    cin>>num;
    cout<<"The final answer is "<<rose(num);

    return 0;
}
```

Without running the code in a text editor or Visual Studio Code, analyze the recursive function algorithm by completing the table below. Assume that the user types 6 from the keyboard:

Call function	rose(int n)	if(n<=0)	else
1	rose(6)	false	rose(5)*6 = 720
2	rose(5)	false	24*5 = 120
3	rose(4)	false	6*4 = 24
4	rose(3)	false	2*3 = 6
5	rose(2)	false	1*2 = 2
6	rose(1)	false	1*1 = 1
7	rose(0)	true	1

What should be the `cout<<"The final answer is "<<rose(num);` ?

Ans : The final answer is 72

Exercise 3: Hypotenuse calculation

Instructions:

1. Open a C++ file and name it using your last name. For example, if your last name is 'Wu', save the file as homework6_Wu.cpp.
2. Add your full name and the homework number as comments at the top of the file as:

```
/*  
Student full name  
Homework 6, functions and introduction to pointers and reference  
*/
```
3. In the C++ file, define the following three functions:
 - **rand_side()**: This function will generate and return a random integer between 1 and 20 (inclusive).
 - **hyp(int side1, int side2)**: This function will calculate and return the hypotenuse of a right-angled triangle using the two integer arguments, side1 and side2. Use the formula:

$$\text{hypotenuse} = \sqrt{\text{side1}^2 + \text{side2}^2}$$

Return the result as a double or float value.

- **print_result(int side1, int side2, double hypotenuse)**: This function will print the two sides and the hypotenuse in the following format:

The hypotenuse of a right angle with side __ and __ is = ____

Where the first blank represents the value of one side, the second blank represents the other side, and the third blank represents the calculated hypotenuse. The hypotenuse should be displayed with two decimal places.

- The **main()** function should ONLY have the following lines and you CANNOT modify the **main()** function.

```
int main(){  
    int side1 = rand_side();  
    int side2 = rand_side();  
    hyp(side1, side2);  
    print_result(side1, side2, hyp);  
    return 0;  
}
```

----- Homework 6 ends here -----